Experiment 16: Compare the Tokenization Outputs of BERT and GPT-2

Python Program

```
from transformers import BertTokenizer, GPT2Tokenizer
bert = BertTokenizer.from_pretrained("bert-base-uncased")
gpt2 = GPT2Tokenizer.from_pretrained("gpt2")
text = "I love learning Python"
print("BERT Tokens:", bert.tokenize(text))
print("BERT IDs:", bert.encode(text, add_special_tokens=False))
print("GPT-2 Tokens:", gpt2.tokenize(text))
print("GPT-2 IDs:", gpt2.encode(text))
```

Output

```
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment16.py"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
BERT Tokens: ['i', 'love', 'learning', 'python']
BERT IDs: [1045, 2293, 4083, 18750]
GPT-2 Tokens: ['I', 'Ġlove', 'Ġlearning', 'ĠPython']
GPT-2 IDs: [40, 1842, 4673, 11361]
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>
```